

Folia: Functional Testing Demo

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Outline

- Problem and vision
- SRS major requirements
- What we built
- Testing strategy and results
- Code coverage
- Bug tracking workflow
- Next steps

Problem and Vision

- Rote word-list study does not build reliable leave intuition
- Drills, repeated practice, and feedback for leave evaluation
- Faster, better decisions through measurable progress, this leads to the mastery of high-frequency leaves

SRS Major Requirements

- Quiz System (QS)
 - constraints/options
 - validation
 - per-question, general session feedback and metrics
- Leave Lookup (LLT)
 - always-available input
 - normalization
 - value display, error fixes
- Nonfunctional including speed, datasets, application path

What We Built (Revamped UI, Added Metrics, Length)

The image shows a web application titled "Folia - Scrabble Leave Value Trainer" in a browser window. The interface is split into two main panels. The left panel, titled "Quiz Mode", contains instructions to click "Start Quiz" to begin, a text input for "Enter your guess:", and two buttons: "Submit Answer" and "Start Quiz". It also displays "Questions: 0/0 | Accurate: 0". The right panel, titled "Leave Lookup", has instructions to "Enter tiles to look up:", a text input, and a "Search" button. A footer bar at the bottom shows session statistics: "Sessions: 1 | Quizzes: 0 | Lifetime Accuracy: 1/1 (100.0%) | Avg Questions/Quiz: 0".

Folia - Scrabble Leave Value Trainer

Quiz Mode

Click 'Start Quiz' to begin

Enter your guess:

Questions: 0/0 | Accurate: 0

Leave Lookup

Enter tiles to look up:

Sessions: 1 | Quizzes: 0 | Lifetime Accuracy: 1/1 (100.0%) | Avg Questions/Quiz: 0

Testing Using PyTest

- Unit
 - LeaveSet (validation, normalization, filters)
 - Quiz/QuizItem (ratings, timing)
- Functional (Frontend)
 - Init, start quiz, guess handling, results, lookup, stats
- Integration
 - Quiz cycles and session statistics

37 different tests

Code Coverage Results

- Overall statements covered: 73%
- QuizItem.py: 100%
- Quiz.py: 92%
- LeaveSet.py: 70%
- scrabble_leave_trainer.py: 67%

Bug Tracking (.md and Github Issues)

☐

Open

8

Closed

0

Author

Labels

Projects

Milestones

Assignees

≡↓ Newest

☐

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Dark Mode, UI Modifications by User

help wanted

#10 · benjaminkramer64 opened now



☐

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Add a Timed Mode

enhancement

#9 · benjaminkramer64 opened now



☐

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Exporting Results

enhancement

#8 · benjaminkramer64 opened 1 minute ago



☐

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Keyboard Shortcuts

enhancement

#7 · benjaminkramer64 opened 2 minutes ago



☐

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Add Difficulty Levels Based on Leaves

enhancement

#6 · benjaminkramer64 opened 2 minutes ago



☐

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Enter Key Fix

bug

#5 · benjaminkramer64 opened 4 minutes ago



☐

•

Window Size Changing

#4 · benjaminkramer64 opened 5 minutes ago

☐

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Accuracy Text Formatting

bug

#3 · benjaminkramer64 opened 5 minutes ago



Next Steps

Increased Coverage (by Nov 15)

- add branch coverage
- expand GUI edge cases

Performance (by Nov 20)

- large CSV load budget, load faster
- make lookups and quiz generation faster

Features (by Nov 28)

- add more metrics and statistics
- add quiz settings (final feature to add)

Appendix

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benjaminkramer@Benjamins-MacBook-Pro-8 src % ./venv/bin/python3.14 -m pytest --cov=. --cov-report=term-missing -q

===== test session starts =====

platform darwin -- Python 3.14.0, pytest-9.0.1, pluggy-1.6.0

rootdir: /Users/benjaminkramer/Desktop/src

configfile: pytest.ini

testpaths: .

plugins: mock-3.15.1, cov-7.0.0

collected 37 items

quiz_test.py [10%]

test_gui_components.py [45%]

test_gui_process.py [64%]

test_integration.py [78%]

test_leave_set.py [100%]

===== tests coverage =====

coverage: platform darwin, python 3.14.0-final-0

Name	Stmts	Miss	Cover	Missing
LeaveSet.py	166	50	70%	23, 25, 29, 39-42, 52, 105, 107-118, 122-126, 128-131, 142, 164, 178-179, 185-186, 191, 194, 197, 206-220
Quiz.py	40	3	92%	18, 45, 49
QuizItem.py	29	0	100%	
scrabble_leave_trainer.py	171	56	67%	30-113, 125-126, 131, 149-150, 173, 234, 240, 268-271, 280-281
TOTAL	406	109	73%	

===== 37 passed in 46.60s =====